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GPT Diffusion Theory

HOW AND WHEN do technological changes affect the rise and fall of great powers? Specifically, how do significant technological breakthroughs result in differential rates of economic growth among great powers? International relations scholars have long observed that rounds of technological revolution lead to upheaval in global economic leadership, bringing about a power transition in the international system. However, few studies explore how this process occurs.

Those that do tend to fixate on the most dramatic aspects of technological change—the eureka moments and first implementations of radical inventions. Consequently, the standard account of technology-driven power transitions stresses a country's ability to dominate innovation in leading sectors. By exploiting brief windows in which to monopolize profits in new industries, the country that dominates innovation in these sectors rises to become the world's most productive economy. Explanations vary regarding why the benefits of leading sectors tend to accrue in certain nations. Some scholars argue that national systems of political economy that accommodate rising challengers can more readily accept and support new industries. Leading economies, by contrast, are victims of their past success, burdened by powerful vested interests that resist adaptation to disruptive technologies.[1] Other studies point to more specific institutional factors that account for why some countries monopolize leading sectors, such as the degree of government centralization or industrial governance structures.[2]

An alternative explanation, based on the diffusion of general-purpose technologies (GPTs), draws attention to the less spectacular process by which

1. Gilpin 1996; Moe 2009.
2. Drezner 2001; Kitschelt 1991.

fundamental innovations gradually diffuse throughout many industries. The rate and scope of diffusion is particularly relevant for GPTs, which are distinguished by their scope for continual improvement, broad applicability across many sectors, and synergies with other technological advances. Recognized by economists and historians as "engines of growth," GPTs hold immense potential for boosting productivity.[3] Realizing this promise, however, necessitates major structural changes across the technology systems linked to the GPT, including complementary innovations, organizational changes, and an upgrading of technical skills. Thus, GPTs lead to a productivity boost only after a "gradual and protracted process of diffusion into widespread use."[4] This is why more than five decades passed before key innovations in electricity, the quintessential GPT, significantly transformed manufacturing productivity.[5]

The process of GPT diffusion illuminates a pathway from technological change to power transition that diverges from the LS account (figure 2.1). Under the GPT mechanism, some great powers sustain economic growth at higher levels than their rivals do because, during a gradual process spanning decades, they more intensively adopt GPTs across a broad range of industries. This is analogous to a marathon run on a wide road. The LS mechanism, in contrast, specifies that one great power rises to economic leadership because it dominates innovations in a limited set of leading sectors and captures the accompanying monopoly profits. This is more like a sprint through a narrow running lane.

Why are some countries more successful at GPT diffusion? Building from scholarship arguing that a nation's success in adapting to emerging technologies is determined by the *fit* between its institutions and the demands of evolving technologies, I argue that the GPT diffusion pathway informs the institutional adaptations crucial to success in technological revolutions.[6] Unlike institutions oriented toward cornering profits in leading sectors, those optimized for GPT diffusion help standardize and spread novel best practices between the GPT sector and application sectors. Education and training systems that widen the base of engineering skills and knowledge linked to new GPTs, or what I call "GPT skill infrastructure," are essential to all of these institutions.

3. Bresnahan and Trajtenberg 1995.
4. David 1990, 356.
5. Devine 1982.
6. Gilpin 1996; Kim and Hart 2001; Kitschelt 1991; Perez 2002.

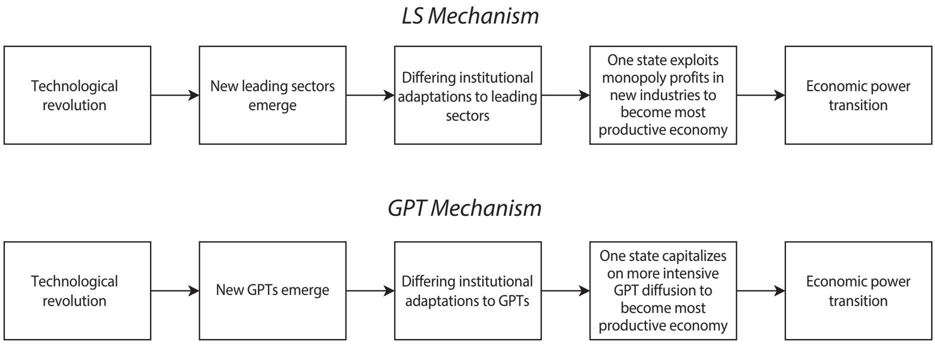

FIGURE 2.1. Causal Diagrams for LS and GPT Mechanisms

The differences between these two theories of technological change and power transition are made clear when one country excels in the institutional competencies for LS product cycles but does not dominate GPT diffusion. Take, for example, chemical innovations and Germany's economic rise in the late nineteenth century. Germany dominated major innovations in chemicals and captured nearly 90 percent of all global exports of synthetic dyestuffs.[7] In line with the LS mechanism, this success was backed by Germany's investments in building R&D labs and training doctoral students in chemistry, as well as a system of industrial organization that facilitated the rise of three chemical giants.[8] Yet it was the United States that held an advantage in adopting basic chemical processes across many industries. As expected by GPT diffusion theory, the United States held institutional advantages in widening the base of engineering skills and knowledge necessary for chemicalization on a wide scale.[9] This is when the ordinary tweakers and the implementers come to the fore, and the star scientists and inventors recede to the background.[10]

The rest of this chapter fleshes out my theoretical framework. It first clarifies the outcome I seek to explain: an economic power transition in which one great power becomes the economic leader by sustaining productivity growth at higher levels than its rivals. The starting point of my argument is that the diffusion of GPTs is central to the relationship between technological change

7. Lehrer 2005, 254–55.
8. Haber 1958, 121–31.
9. Rosenberg and Steinmueller 2013. A later section of this chapter expands on U.S. advantages in chemical engineering.
10. I adopt the terms "tweaker" and "implementer" from Meisenzahl and Mokyr 2011, 446.

and productivity leadership. This chapter explicates this argument by justifying the emphasis on both GPTs and diffusion, highlighting the differences between the GPT and LS mechanisms. It then extends the analysis to the institutional competencies that synergize with GPT trajectories. From the rich set of technology-institution interactions identified by evolutionary economists and comparative institutionalists, I justify my focus on institutions that enable countries to widen the skill base required to spread GPTs across industries. After differentiating my argument from alternative explanations, the chapter closes with a description of the book's research methodology.[11]

The Outcome: Long-Term Economic Growth Differentials and Power Transitions

Power transitions are to the international system as earthquakes are to the geological landscape. Shifts in the relative power of leading nations send shock waves throughout the international system. What often follows is conflict, the most devastating form of which is a war waged by coalitions of great powers for hegemony over the globe.[12] Beyond heightened risks of conflict, the aftershocks of power transitions reverberate in the architecture of the international order as victorious powers remake international institutions in their own images.[13]

While the power transition literature largely tackles the consequences of power transitions, I treat the rise and fall of great powers as the outcome to be explained. This follows David Baldwin's instruction for international relations scholars "to devote more attention to treating power as a dependent variable and less to treating it as an independent variable."[14] Specifically, I explore the causes of "economic power transitions," in which one great power sustains economic growth rates at higher levels than its rivals.[15]

11. Parts of this chapter derive from Ding 2024.

12. Power transition theory expects that the risk of a major war is greatest when a rising challenger threatens the established power (Gilpin 1975; Organski 1958; Tammen 2008). For a critique, see Chan 2007.

13. For discussions of other possible consequences of power shifts, such as discussions of hegemonic stability, see Keohane 1984; Snidal 1985.

14. Baldwin 2012, 288.

15. My outcome variable, differential rates of economic growth among established and rising powers, is regarded by international relations scholars as a key stage in the overarching process of power transition (Gilpin 1981). Related literature references shifts in industrial leadership (Moe 2009), leading economies (Modelski and Thompson 1996; Reuveny and Thompson

It might not be obvious, at first glance, why I focus on economic power. After all, power is a multidimensional, contested concept that comes in many other forms. The salience of certain power resources depends on the context in which a country draws upon them to exert influence.[16] For my purposes, differentials in economic growth are the most relevant considerations for intensifying hegemonic rivalry. An extensive literature has demonstrated that changes in relative economic growth often precede hegemonic wars.[17]

Moreover, changes in global political and military leadership often follow shifts in economic leadership. As the most fungible mode of power, economic strength undergirds a nation's influence in global politics and its military capabilities.[18] The outcomes of interstate conflicts bear out that economic and productive capacity is the foundation of military power.[19] Paul Kennedy concludes that

> all of the major shifts in the world's military-power balances have followed alterations in the productive balances . . . the rising and falling of the various empires and states in the international system has been confirmed by the outcomes of the major Great Power wars, where victory has always gone to the side with the greatest material resources.[20]

How does one identify if or when an economic power transition occurs? Phrased differently, how many years does a great power need to lead its rivals in economic growth rates? How large does that gap have to reach? Throughout this book, I judge whether an economic power transition has occurred based on one great power attaining a lead in overall economic productivity over its

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2001), and the technological hegemon (Drezner 2001, 4). I use the term "economic power transition" because it concisely captures the outcome I focus on while avoiding the associations of "industrial" with heavy industry.

16. For instance, a large military may be an especially important component of a state's influence in fighting conventional wars but not as salient for that state's influence in defending against cyberattacks from nonstate actors. David Baldwin (2016) cautions against making context-free estimates of power resources and reducing the multidimensional concept of power into a single measure.

17. Kennedy 1987; Kim and Morrow 1992; Kugler and Lemke 1996; Väyrynen 1983. For a view that complicates the relationship between differential rates of economic growth and war, see Debs and Monteiro 2014.

18. Huntington 1993, 81; see also Monteiro 2014, 34.

19. Kirshner 1998; Modelski and Thompson 1996. On World War I, see Kennedy 1976; on World War II, see Hanson 2017. Economic strength does not map perfectly onto military strengths. Some economic giants, such as Japan in the 1980s, limit their military capabilities.

20. Kennedy 1987, 439.

rivals by sustaining higher levels of productivity growth rates.[21] Productivity growth ensures that efficient and sustainable processes are fueling growth in total economic output. Additionally, productivity is the most important determinant of economic growth in the long run, which is the appropriate time horizon for understanding power transitions. "Productivity isn't everything, but in the long run it is almost everything," states Nobel Prize–winning economist Paul Krugman.[22]

Alternative conceptualizations of economic power cannot capture how effectively a country translates technological advance into national economic growth. Theories of geo-economics, for instance, highlight a state's balance of trade in certain technologically advanced industries.[23] Other studies emphasize a state's share of world-leading firms.[24] National rates of innovation, while more inclusive, measure the generation of novel technologies but not diffusion across commercial applications, thereby neglecting the ultimate impact of technological change.[25] Compared to these indicators, which account for only a small portion of the value-added activities in the economy, productivity provides a more comprehensive measure of economic leadership.[26]

This focus on productivity is supported by recent work on power measurement, which questions measures of power resources based on economic size. Without accounting for economic efficiency, solely relying on measures of gross economic and industrial output provides a distorted view of the balance of power, particularly where one side is populous but poor.[27] If national power was measured by GDP alone, China was the world's most powerful country during the first industrial revolution. However, China's economy was far from the productivity frontier. In fact, as the introduction chapter spotlighted, the view that China fell behind the West because it

21. Throughout the book, to vary word choice, I describe the outcome variable as a shift in "productivity leadership," as an "economic power transition," or as change to the "economic balance of power." These all refer to the same concept.

22. Krugman 1997, 11.

23. Kim 2020; Luttwak 1993.

24. Sean Starrs (2013) measures American economic power by the profit shares of US-headquartered transnational corporations.

25. Taylor 2012, 2016.

26. Krugman 1994. Most international political economy scholars regard productivity growth as the most telling measure of technological competitiveness (Hart 1992, 15; Porter 1990, 5).

27. Anders, Fariss, and Markowitz 2020; Beckley 2018.

could not capitalize on productivity-boosting technological breakthroughs is firmly entrenched in the minds of leading Chinese policymakers and thinkers.

Lastly, it is important to clarify that I limit my analysis of productivity differentials to great powers.[28] In some measures of productivity, other countries may rank highly or even outrank the countries I study in my cases. In the current period, Switzerland and other countries have higher GDP per capita than the United States; before World War I, Australia was the world leader in productivity, as measured by GDP per labor-hour.[29] However, smaller powers like pre–World War I Australia and present-day Switzerland are excluded from my study of economic power transitions, as they lack the baseline economic and population size to be great powers.[30]

There is no exact line that distinguishes great powers from other countries.[31] Kennedy's seminal text *The Rise and Fall of the Great Powers*, for instance, has been challenged for not providing a precise definition of great power.[32] Fortunately, across all the case studies in this book, there is substantial agreement on the great powers of the period. According to one measure of the distribution of power resources, which spans 1816 to 2012 and incorporates both economic size and efficiency, all the countries I study rank among the top six at the beginning of the case.[33]

The Diffusion of GPTs

Scholars often gravitate to technological change as the source of a power transition in which the mantle of industrial preeminence changes hands. However, there is less clarity over the process by which technical breakthroughs translate

28. I am indebted to Allan Dafoe, Max Daniel, and Duncan Snidal for helping me clarify the points that follow.

29. Wright 1990, 653.

30. Australia's population in 1870 was 1.6 million, which was only 4 percent of that of the United States at the time (Romer 1996, 202).

31. Some definitions of "great powers" regard powerful military capabilities as a requirement (Monteiro 2014, 44). Kennedy (1987, 224) writes that a great power is "by definition, a state capable of holding its own against any other nation." I do not make military capabilities a necessary criterion for great powers in my study because some rising powers become economic leaders before they become capable of projecting military power overseas. In work in progress, Jennifer Lind (2023) employs the term "latent great powers" to characterize these powers. I am grateful to her for sharing insights on this topic. See also Mearsheimer 2014.

32. D. Baldwin 2016, 103; Kaiser 1989, 738.

33. My calculations are based on data from Anders, Fariss, and Markowitz 2020.

into this power shift among countries at the technological frontier. I argue that the diffusion of GPTs is the key to this mechanism. In this section, I first outline why my theory privileges GPTs over other types of technology. I then set forth why diffusion should be prioritized over other phases of technological change, especially innovation. Finally, I position GPT diffusion theory against the leading sector (LS) model, which is the standard explanation in the international relations literature.

Why GPTs?

Not all technologies are created equal. When assessed on their potential to transform the productivity of nations, some technical advances, such as the electric dynamo, rank higher than others, such as an improved sleeping bag. My theory gives pride of place to GPTs, such as electricity and the steam engine, which have historically generated waves of economy-wide productivity growth.[34] Assessed on their own merits alone, even the most transformative technological changes do not tip the scale far enough to significantly affect aggregate economic productivity.[35] GPTs are different because their impact on productivity comes from accumulated improvements across a wide range of complementary sectors; that is, they cannot be judged on their own merits alone.

Recognized by economists and economic historians as "engines of growth," GPTs are defined by three characteristics.[36] First, they offer *great potential for continual improvement*. While all technologies offer some scope for improvement, a GPT "has implicit in it a major research program for improvements, adaptations, and modifications."[37] Second, GPTs acquire *pervasiveness*. As a GPT evolves, it finds a "wide variety of uses" and a "wide range of uses."[38] The former refers to the diversity of a GPT's use cases, while the latter alludes to

34. Brynjolfsson, Rock, and Syverson 2017; David 1990; Ruttan 2006, 5.

35. For instance, Robert William Fogel's classic study of railroads concluded that "the railroad did not make an overwhelming contribution to the production potential of the economy" (Fogel 1964, 235). Related categories, including "enabling technologies," spur complementary innovations and exhibit substantial scope for improvement but do not find pervasive applications (Teece 2018, 1369).

36. Bresnahan and Trajtenberg 1995. The following discussion is mostly drawn from Bresnahan and Trajtenberg 1995 and Lipsey, Carlaw, and Bekar 2005. Other accounts employ similar definitions, albeit with some modifications; see Bresnahan 2010; Jovanovic and Rousseau 2005. For a critical view of the GPT concept, see Field 2008.

37. Lipsey, Bekar, and Carlaw 1998, 39.

38. Cantner and Vannuccini 2012.

the breadth of industries and individuals using a GPT.[39] Third, GPTs have *strong technological complementarities*. In other words, the benefits from innovations in GPTs come from how other linked technologies are changed in response and cannot be modeled from a mere reduction in the costs of inputs to the existing production function. For example, the overall energy efficiency gains from merely replacing a steam engine with an electric motor were minimal; the major benefits from factory electrification came from electric "unit drive," which enabled machines to be driven individually by electric motors, and a radical redesign of plants.[40]

Taken together, these characteristics suggest that the full impact of a GPT materializes via an "extended trajectory" that differs from those associated with other technologies. Economic historian Paul David explains:

> We can recognize the emergence of an extended trajectory of incremental technical improvements, the gradual and protracted process of diffusion into widespread use, and the confluence with other streams of technological innovation, all of which are interdependent features of the dynamic process through which a general purpose engine acquires a broad domain of specific applications.[41]

For example, the first dynamo for industrial application was introduced in the 1870s, but the major boost of electricity to overall manufacturing productivity did not occur until the 1920s. Like other GPT trajectories, electrification required a protracted process of workforce skill adjustments, organizational adaptations, such as changes in factory layout, and complementary innovations like the steam turbine, which enabled central power generation in the form of utilities.[42] To track the full impact of these engines of growth, one must travel the long roads of their diffusion.

Why Diffusion?

All technological trajectories can be divided into a phase when the technology is incubated and then first introduced as a viable commercial application ("innovation") and a phase when the innovation spreads through a population of

39. One does not imply the other. For instance, a screw has a "wide range of use" since it is used to fasten things together across a large swath of productivity activities in the economy, but it does not have a "wide variety of uses" (Lipsey, Bekar, and Carlaw 1998, 39).

40. Devine 1982.

41. David 1990, 356.

42. Brynjolfsson, Rock, and Syverson 2017; David 1990; Smil 2005, 33–97.

potential users, both nationally and internationally ("diffusion").[43] Recognizing this commonly accepted distinction, other studies of the scientific and technological capabilities of nations primarily focus on innovation.[44] I depart from other works by giving priority to diffusion, since that is the phase of technological change most significant for GPTs.[45]

Undeniably, the activities and conditions that produce innovation can also spur diffusion.[46] A firm's ability to conduct breakthrough R&D does not just create new knowledge but also boosts its capacity to assimilate innovations from external sources ("absorptive capacity").[47] Faced with an ever-shifting technological frontier, building competency at producing new innovations gives a firm the requisite prior knowledge for identifying and commercializing external innovations. Other studies extend these insights beyond firms to regional and national systems.[48] In order to absorb and diffuse technological advances first incubated elsewhere, they argue, nations must invest in a certain level of innovative activities.

This connection between innovation capacity and absorptive capacity could question the GPT mechanism's attention to diffusion. Possibly, a country's lead in GPT innovation could also translate directly into a relative advantage in GPT diffusion.[49] Scholarship on the agglomeration benefits of

43. "GPT diffusion" refers to the assimilation and adoption of GPTs *within* one country's economic system. A rich scholarship on diffusion studies how norms and policies spread *across* countries (Finnemore and Sikkink 1998; Simmons and Elkins 2004). When discussing how GPTs spread from one economy to another, I use the term "international diffusion."

44. Existing work on the scientific and technological power of nations makes the same distinction. In his book, Andrew Kennedy (2018, 16) writes, "I focus on transnational processes that support firms' and universities' efforts to engage in the first of these tasks: the creation of a product or process that is 'new to the world.'" Mark Zachary Taylor (2016, 28, emphasis in original) writes that his book, *The Politics of Innovation*, is more interested in "*innovation* than *diffusion* . . . where possible, I focus more on why some countries are better at inventing new technologies. . . . I am somewhat less concerned with the spread of new technology throughout society." See also Taylor 2009, 865.

45. A diffusion-oriented approach checks the "innovation-centrism" that has permeated the analysis of technology in the social sciences (Edgerton 2010, 689). In his book-length treatment of the concept of innovation, Benoît Godin (2015, 8) describes the current moment as one when "innovation becomes a value per se" and an "object of veneration and cult worship." Relatedly, studies of technological change in military affairs are strongly biased toward military innovation (Goldman and Andres 1999, 80n4).

46. Taylor 2016, 231. Joel Simmons (2016, 33) has argued that "the differences between the two processes often are exaggerated."

47. Aghion and Howitt 1998; Cohen and Levinthal 1998.

48. Fagerberg 1987; Fu 2008; Howitt and Mayer-Foulkes 2005.

49. Arnulf Grübler (1998), for example, argues that the leading innovation center, the country where new technological changes originate, tends to also have the highest adoption levels of the technology.

innovation hot spots, such as Silicon Valley, support this case to some extent. Empirical analyses of patent citations indicate that knowledge spillovers from GPTs tend to cluster within a geographic region.[50] In the case of electricity, Robert Fox and Anna Guagnini underscore that it was easier for countries with firms at the forefront of electrical innovation to embrace electric power at scale. The interconnections between the "learning by doing" gained on the job in these leading firms and academic labs separated nations in the "fast lane" and "slow lane" of electrification.[51]

Being the first to pioneer new technologies could benefit a state's capacity to absorb and diffuse GPTs, but it is not determinative. A country's absorptive capacity also depends on many other factors, including institutions for technology transfer, human capital endowments, openness to trade, and information and communication infrastructure.[52] Sometimes the "advantages of backwardness" allow laggard states to adopt new technologies faster than the states that pioneer such advances.[53] In theory and practice, a country's ability to generate fundamental, new-to-the-world innovations can widely diverge from its ability to diffuse such advances.

This potential divergence is especially relevant for advanced economies, which encompass the great powers that are the subject of this research. Although innovation-centered explanations do well at sorting the advantages of technological breakthroughs to countries at the technological frontier compared to those trying to catch up, they are less effective at differentiating among advanced economies. As supported by a wealth of econometric research, divergences in the long-term economic growth of countries at the technology frontier are shaped more by *imitation* than innovation.[54] These advanced countries have firms that can quickly copy or license innovations; first mover advantages from innovations are thus limited even in industries, like pharmaceuticals, that enforce intellectual property rights most strictly.[55] Nevertheless, advanced countries that are evenly matched in their capacity for radical innovation can undertake vastly different growth trajectories in the wake of technological revolutions. Differences in diffusion pathways are central to explaining this puzzle.

50. Feldman and Yoon 2012.
51. Fox and Guagnini 2004, 170.
52. Comin and Hobijn 2010; Fu 2008.
53. Gerschenkron 1962/2008.
54. Fagerberg 1987; Pavitt and Soete 1982.
55. Hannah 1994, 90–91. I thank Daniel Raff for sharing this text with me.

This diffusion-centered approach is especially well suited for GPTs. Since GPTs entail gradual evolution into widespread use, there is a longer window for competitors to adopt GPTs more intensively than the leading innovation center. In other technologies, first-mover benefits from pioneering initial breakthroughs are more significant. For instance, leadership in the innovation of electric power technologies was fiercely contested among the industrial powers. The United States, Germany, Great Britain, and France all built their first central power stations within a span of three years (1882–1884), their first electric trams within a span of nine years (1887–1896), and their first three-phase AC power systems within a span of eight years (1891–1899).[56] However, the United States clearly led in the diffusion of these systems: by 1912, its electricity production per capita had more than doubled that of Germany, its closest competitor.[57] Thus, while most countries at the technological frontier will be able to compete in the production and innovation of GPTs, the hardest hurdles in the GPT trajectory are in the diffusion phase.

GPT Diffusion and LS Product Cycles

GPT diffusion challenges the LS-based account of how technological change drives power transitions. The standard explanation in the international relations literature emphasizes a country's dominance in leading sectors, defined as new industries that experience rapid growth on the back of new technologies.[58] Cotton textiles, steel, chemicals, and the automobile industry form a "classic sequence" of "great leading sectors," developed initially by economist Walt Rostow and later adapted by political scientists.[59] Under the LS mechanism, a country's ability to maintain a monopoly on innovation in these emerging industries determines the rise and fall of lead economies.[60]

This model of technological change and power transition builds on the international product life cycle, a concept pioneered by Raymond Vernon. Constructed to explain patterns of international trade, the cycle begins with a product innovation and subsequent sales growth in the domestic market.

56. Taylor 2016, 189.
57. Author's calculations based on Comin and Hobijn 2009.
58. Kennedy 2018, 51; Rostow 1960, 14.
59. Rostow 1978, 104–9. Thompson (1990) uses ten indicators proposed by Rostow as proxies for the rise and fall of critical leading sectors, adding two new indicators for semiconductor and jet airframe production starting in the 1950s.
60. Thompson 1990, 217.

Once the domestic market is saturated, the new product is exported to foreign markets. Over time, production shifts to these markets, as the original innovating country loses its comparative advantage.[61]

LS-based studies frequently invoke the product cycle model.[62] Analyzing the effects of leading sectors on the structure of the international system, Gilpin states, "Every state, rightly, or wrongly, wants to be as close as possible to the innovative end of 'the product cycle.'"[63] One scholar described Gilpin's *US Power and the Multinational Corporation*, one of the first texts that outlines the LS mechanism, as "[having] drawn on the concept of the product cycle, expanded it into the concept of the growth and decline of entire national economies, and analyzed the relations between this economic cycle, national power, and international politics."[64]

The product cycle's assumptions illuminate the differences between the GPT and LS mechanisms along three key dimensions. In the first stage of the product cycle, a firm generates the initial product innovation and profits from sales in the domestic market before saturation. Extending this model to national economies, the LS mechanism emphasizes the clustering of LS innovations and attendant monopoly profits in a single nation.[65] "The extent of national success that we have in mind is of the fairly extreme sort," write George Modelski and William Thompson. "One national economy literally dominates the leading sector during its phase of high growth and is the primary beneficiary of the immediate profits."[66] The GPT trajectory, in contrast, places more value on where technologies are diffused than where an innovation is first pioneered.[67] I refer to this dimension as the "phase of relative advantage."

In the next stage, the product innovation spreads to global markets and the technology gradually diffuses to foreign competitors. Monopoly rents associated with a product innovation dissipate as production becomes routinized

61. Vernon 1971.

62. Gilpin 1975, 78, 197; 1987, 234–37; Moe 2007, 207; Tellis et al. 2000, 37.

63. Gilpin 1987, 99.

64. Kurth 1979, 4.

65. Rasler and Thompson 1994, 7.

66. Modelski and Thompson 1996, 91; see also Thompson 1990, 217. It is important to note that recent work on technology and systemic leadership points out that, starting during the postwar period, it is more difficult for one state to completely monopolize innovation. See Thompson 2022, 184–211.

67. For other political science work that emphasizes variation in technological adoption, see Milner 2006; Milner and Solstad 2021.

and transfers to other countries.[68] Mirroring this logic, Modelski and Thompson write, "[Leading sectors] bestow the benefits of monopoly profits on the pioneer until diffusion and imitation transform industries that were once considered radically innovative into fairly routine and widespread components of the world economy."[69] Thompson also states that "the sector's impact on growth tends to be disproportionate in its early stages of development."[70]

The GPT trajectory assumes a different impact timeframe. The more wide-ranging the potential applications of a technology, the longer the lag between its initial emergence and its ultimate economic impact. This explains why the envisioned transformative impact of GPTs does not appear straightaway in the productivity statistics.[71] Time for complementary innovations, organizational restructuring, and institutional adjustments such as human capital formation is needed before the full impact of a GPT can be known. It is precisely the period when diffusion transforms radical innovations into routine components of the economy—the stage when the causal effects of leading sectors are expected to dissipate—that generates the productivity gap between nations.

The product cycle also reveals differences between the LS and GPT mechanisms regarding the "breadth of growth." Like the product cycle's focus on an innovation's life cycle within a singular industry, the LS mechanism emphasizes the contributions of a limited number of new industries to economic growth in a particular period. GPT-fueled productivity growth, on the other hand, is dispersed across a broad range of industries.[72] Table 2.1 specifies how LS product cycles differ from GPT diffusion along the three dimensions outlined here. As the following section will show, the differences in these two technological trajectories shape the institutional factors that are most important for national success in adapting to periods of technological revolution.

68. "Monopoly rents" refers to a condition wherein a producer lacks competition and can sell its goods and services at an above-market price. In the context of LS theory, it is the lead economy that monopolizes rents from new technologies, in the sense that it accumulates higher profits from a fast-growing industry. I thank an anonymous reviewer for pushing me to clarify this point.

69. Modelski and Thompson 1996, 52.

70. Thompson 1990, 211. The impact timeframe of leading sectors is like "distributing money on the ground," write Abraham Newman and John Zysman (2006, 393–94). "Some radically valuable possibilities, the larger bills, are picked up first; the smaller opportunities are captured later. But the original technological revolution loses force, as the most valuable opportunities are picked up and implemented."

71. Brynjolfsson, Rock, and Syverson 2017; David 1990; Helpman and Trajtenberg 1994.

72. Crafts 2001, 306; David and Wright 1999, 12.

TABLE 2.1. Two Mechanisms of Technological Change and Power Transitions

Mechanisms	Impact Timeframe	Phase of Relative Advantage	Breadth of Growth	Institutional Complements
LS product cycles	Lopsided in early stages	Monopoly on innovation	Concentrated	Deepen skill base in LS innovations
GPT diffusion	Lopsided in later stages	Edge in diffusion	Dispersed	Widen skill base in spreading GPTs

While I have highlighted the differences between GPT diffusion and LS product cycles, it is important to recognize that there are similarities between the two pathways.[73] Some scholars, for example, associate leading sectors with broad spillovers across economic sectors.[74] In addition, lists of leading sectors and lists of GPTs sometimes overlap, as evidenced by the fact that electricity is a consensus inclusion on both lists. Moreover, both explanations begin with the same premise: to fully uncover the dynamics of technology-driven power transitions, it is essential to specify which new technologies are the key drivers of economic growth in a particular time window.[75]

At the same time, these resemblances should not be overstated. Many classic leading sectors do not have general-purpose applications. For instance, cotton textiles and automobiles both feature on Rostow's series of leading sectors, and they are studied as leading sectors because each has "been the largest industry for several major industrial nations in the West at one time or another."[76] Although these were certainly both fast-growing large industries, the underlying technological advances do not fulfill the characteristics of GPTs. In addition, many of the GPTs I examine do not qualify as leading sectors. The machine tool industry in the mid-nineteenth century, for instance, was not a new industry, and it was never even close to being the largest industry in any of the major economies. Most importantly, though the GPT and LS

73. Gilpin 1987, 217; Reuveny and Thompson 2001, 711n13.

74. Drezner 2001, 7; Thompson 1990, 211.

75. I share the opinion of one of the earliest LS-based texts that "revolutionary technological breakthroughs" are the "primary concern in explaining the rise and decline of successive capitalistic core economies" (Gilpin 1975, 69). I differ on which technologies are considered "revolutionary" and on how these breakthroughs actually resulted in the rise and decline of major economies.

76. Kurth 1979, 3.

mechanisms sometimes point to similar technological changes, they present very different understandings of *how* revolutionary technologies bring about an economic power transition. As the next section reveals, these differences also map onto varied institutional adaptations.

GPT Skill Infrastructure

New technologies agitate existing institutional patterns.[77] They appeal for government support, generate new collective interests in the form of technical societies, and induce organizations to train people in relevant fields. If institutional environments are slow or fail to adapt, the development of emerging technologies is hindered. As Gilpin articulates, a nation's technological fitness is rooted in the "extent of the congruence" between its institutions and the demands of evolving technologies.[78] This approach is rooted in a rich tradition of work on the coevolution of technology and institutions.[79]

Understanding the demands of GPTs helps filter which institutional factors are most salient for how technological revolutions bring about economic power transitions. Which institutional factors dictate disparities in GPT adoption among great powers? Specifically, I emphasize the role of education and training systems that broaden the base of engineering skills linked to a particular GPT. This set of institutions, which I call "GPT skill infrastructure," is most crucial for facilitating the widespread adoption of a GPT.

To be sure, GPT diffusion is dependent on institutional adjustments beyond GPT skill infrastructure. Intellectual property regimes, industrial relations, financial institutions, and other institutional factors could affect GPT diffusion. Probing inter-industry differences in technology adoption, some studies find that less concentrated industry structures are positively linked to GPT adoption.[80] I limit my analysis to institutions of skill formation because their effects permeate other institutional arrangements.[81] GPT skill infrastruc-

77. I am very thankful for feedback from Stephen Kaplan and Abe Newman on this section.

78. Gilpin 1996, 413.

79. Freeman and Louca 2001; Nelson and Winter 1982; Perez 2002.

80. Romeo 1975; Copeland and Shapiro 2010.

81. Thelen 2004, 4, 285–86. A substantial body of literature stresses the importance of human capital for technology adoption; see, for example, Comin and Hobijn 2004; Goldin and Katz 2008; Nelson and Phelps 1966. Later in the chapter, I outline my tests for the validity of two alternative explanations for cross-national differences in GPT diffusion.

ture provides a useful indicator for other institutions that standardize and spread the novel best practices associated with GPTs.[82]

It should also be noted that the institutional approach is one of three main categories of explanation for cross-country differences in economic performance over the long term.[83] Other studies document the importance of geography and culture to persistent cross-country income differences.[84] I prioritize institutional explanations for two reasons. First, natural experiments from certain historical settings, in which institutional divergence occurs but geographical and cultural factors are held constant, suggest that institutional differences are particularly influential sources of long-term economic growth differentials.[85] Second, since LS-based accounts of power transitions also prioritize institutional adaptations to technological change, my approach provides a more level test of GPT diffusion against the standard explanation.[86]

One final note about limits to my argument's scope. I do not investigate the deeper origins of why some countries are more effective than others at developing GPT skill infrastructure. Possibly, the intensity of political competition and the inclusiveness of political institutions influence the development of skill formation institutions.[87] Other fruitful lines of inquiry stress the importance of government capacity to make intertemporal bargains and adopt long time horizons in making technology investments.[88] It is worth noting that a necessary first step to productively exploring these underlying causes is to establish which types of technological trajectories and institutional adaptations are at work. For instance, LS product cycles may be more closely linked to mercantilist or state capitalist approaches that favor narrow interest groups, whereas, political systems that incorporate a broader group of stakeholders may better accommodate GPT diffusion pathways.

82. Rosenberg 1998b; Vona and Consoli 2014.

83. For a review of these "three fundamental causes" of international patterns of growth, see Acemoglu, Johnson, and Robinson 2005, 397–402.

84. For geographical explanations, see Diamond 1997; Gallup, Sachs, and Mellinger 1999. For cultural theories, see Harrison and Huntington 2000; Weber 1930.

85. Acemoglu, Johnson, and Robinson 2005, 402–21.

86. Some scholarship does explore how certain geographical settings are more conducive to specific types of technological change, but these studies are limited to agricultural technologies. See, for example, Diamond 1997, 358.

87. Acemoglu and Robinson 2006, 2012.

88. Doner and Schneider 2016; Simmons 2016; see also Doner, Hicken, and Ritchie 2009.

Institutions Fit for GPT Diffusion

If GPTs drive economic power transitions, which institutions fit best with their demands? Institutional adaptations for GPT diffusion must solve two problems. First, since the economic benefits of GPTs materialize through improvements across a broad range of industries, capturing these benefits requires extensive coordination between the GPT sector and numerous application sectors. Given the sheer scope of potential applications, it is infeasible for firms in the GPT sector to commercialize the technology on their own, as the necessary complementary assets are embedded with different firms and industries.[89] In the AI domain, as one example of a potential GPT, firms that develop general machine learning algorithms will not have access to all the industry-specific data needed to fine-tune those algorithms to particular application scenarios. Thus, coordination between the GPT sector and other organizations that provide complementary capital and skills, such as academia and competitor firms, is crucial. In contrast, for technologies that are *not* general-purpose, this type of coordination is less conducive and could even be detrimental to a nation's competitive advantage, as the innovating firm could leak its technical secrets.[90]

Second, GPTs pose demanding conditions for human capital adjustments. In describing the connection between skill formation and technological fitness, scholars often delineate between general skills and industry-specific skills. According to this perspective, skill formation institutions that optimize for the former are more conducive to technological domains characterized by radical innovation, while institutions that optimize for the latter are more favorable for domains marked by incremental innovation.[91] GPT diffusion entails both types of skill formation. The skills must be specific to a rapidly changing GPT domain but also broad enough to enable a GPT's advance across many industries.[92] Strong linkages between R&D-intensive organizations at the technological frontier and application areas far from the frontier also play a key role in GPT diffusion. This draws attention to the interactions between researchers who produce innovations and technicians who help absorb them into specific contexts.[93]

89. Teece 2018.
90. Goldfarb, Taska, and Teodoridis 2021.
91. Hall and Soskice 2001; Culpepper and Finegold 1999.
92. Aghion and Howitt 2002, 312–13; Streeck 1992, 16.
93. Mason, Rincon-Aznar, and Venturini 2020; Rincon-Aznar et al. 2015.

Education and training systems that foster relevant engineering skills for a GPT, or what I call GPT skill infrastructure, address both constraints. Engineering talent fulfills the need for skills that are rooted in a GPT yet sufficiently flexible to implement GPT advances in a wide range of sectors. Broadening the base of engineering knowledge also helps standardize best practices with GPTs, thereby coordinating information flows between the GPT sector and application sectors. Standardization fosters GPT diffusion by committing application sectors to specific technological trajectories and encouraging complementary innovations.[94] This unlocks the horizontal spillovers associated with GPTs.[95]

Indeed, distinct engineering specializations have emerged in the wake of a new GPT. New disciplines, such as chemical engineering and electrical engineering, have proved essential in widening knowledge bases in the wake of a new GPT.[96] Computer science, another engineering-oriented field, was central to US leadership in the information revolution.[97] These professions developed alongside new technical societies—ranging from the American Society of Mechanical Engineers to the Internet Engineering Task Force—that formulated and disseminated guidelines and benchmarks for GPT development.[98]

Clearly, the features of GPT skill infrastructure have changed over time. Whereas informal associations systematized the skills crucial for mechanization in the eighteenth century, formal higher education has become increasingly integral to computerization in the twenty-first century.[99] Some evidence suggests that computers and other technologies are skill-biased, in the sense that they favor workers with more years of schooling.[100] These

94. Baron and Schmidt 2014; Bresnahan and Trajtenberg 1995; Thoma 2009.

95. Because it is more difficult for the GPT sector to capture all associated spillovers, relative to industries connected to narrower "discrete" technologies, the GPT sector may underinvest in innovation (Bresnahan 2010; Bresnahan and Trajtenberg 1995; Gambardella et al. 2021). I do not concentrate as much on this appropriability problem because great powers are likely to have access to a source of GPT development. What differentiates them is the breadth and depth of commercialization.

96. Rosenberg 1998b, 169. Historical studies attribute the successful adoption of new electrical technologies in the United States to the "better match between the technologies advanced by electrification and the country's institutions of education and worker training" (David and Wright 2006, 154).

97. Vona and Consoli 2014, 1403–5.

98. Russell 2014; Yates and Murphy 2019.

99. Still, recent studies emphasize the continuing importance of informal education opportunities for computer science (Guzdial et al. 2014).

100. Johnson 1997; Krueger 1993.

trends complicate but do not undercut the concept of GPT skill infrastructure. Regardless of the extent of formal training, all configurations of GPT skill infrastructure perform the same function: to widen the pool of engineering skills and knowledge associated with a GPT. This can take place in universities as well as in informal associations, provided these settings train engineers and facilitate the flow of engineering knowledge between knowledge-creation centers and application sectors.[101]

Which Institutions Matter?

The institutional competencies for exploiting LS product cycles are different. Historical analysis informed by this frame highlights heroic inventors like James Watt and pioneering research labs at large companies.[102] Studying which countries benefited most from emerging technologies over the past two centuries, Herbert Kitschelt prioritizes the match between the properties of new technologies and sectoral governance structures. Under his framework, for example, tightly coupled technological systems with high causal complexity, such as nuclear power systems and aerospace platforms, are more likely to flourish in countries that allow for extensive state support.[103] In other studies, the key institutional factors behind success in LS product cycles are education systems that subsidize scientific training and R&D facilities in new industries.[104]

These approaches equate technological leadership with a state's success in capturing market shares and monopoly profits in new industries.[105] In short, they use LS product cycles as the filter for which institutional variables matter. Existing scholarship lacks an institutional explanation for why some great powers are more successful at GPT diffusion.

Competing interpretations of technological leadership in chemicals during the late nineteenth century crystallize these differences. Based on the LS template, the standard account accredits Germany's dominance in the chemical

101. Mokyr and Voth 2010.
102. See, for example, Kennedy 2018, 54.
103. Kitschelt 1991; see also Kim and Hart 2001.
104. Drezner 2001; Moe 2009, 216–17.
105. Another example of this is Espen Moe's work on the industrial rise and fall of great powers, which takes a state's competitiveness in a particular period's "core industry" to be equivalent to a state's industrial leadership. His choice of core industries draws from leading sectors identified in Gilpin 1987 and Modelski and Thompson 1996 (Moe 2009, 224fn3; Moe 2007).

industry—as represented by its control over 90 percent of global production of synthetic dyes—to its investments in scientific research and highly skilled chemists.[106] Germany's dynamism in this leading sector is taken to explain its overall industrial dominance.[107]

GPT diffusion spotlights a different relationship between technological change and institutional adaptation. The focus turns toward institutions that complemented the extension of chemical processes to a wide range of industries beyond synthetic dye, such as food production, metals, and textiles. Under the GPT mechanism, the United States, not Germany, achieved leadership in chemicals because it first institutionalized chemical engineering as a discipline. Despite its disadvantages in synthetic dye production and chemical research, the United States was more effective in broadening the base of chemical engineering talent and coordinating information flows between fundamental breakthroughs and industrial applications.[108]

It is important to note that some parts of the GPT and LS mechanisms can coexist without conflict. A state's capacity to pioneer new technologies can correlate with its capacity to absorb and diffuse GPTs. Countries that are home to cutting-edge R&D infrastructure may also be fertile ground for education systems that widen the pool of GPT-linked engineering skills. However, these aspects of the LS mechanism are not necessary for the GPT mechanism to operate. In accordance with GPT diffusion theory, a state can capitalize on GPTs to become the most powerful economy without monopolizing LS innovation.

Moreover, other dimensions of these two mechanisms directly conflict. When it comes to impact timeframe and breadth of growth, the GPT and LS mechanisms advance opposing expectations. Institutions suited for GPT diffusion can diverge from those optimized for creating new-to-the-world innovations. Research on human capital and long-term growth separates the effects of engineering capacity, which is commonly tied to adoptive activities, and other forms of human capital that are more often connected to inventive activities.[109] This divergence can also be seen in debates over the effects of competition on technological activity. On the one hand, Joseph Schumpeter and others have argued that monopoly structures incentivize more R&D activity because the

106. Drezner 2001, 13–18; Moe 2007, 125.
107. Moe 2007, 253–55; Thompson 1990.
108. Rosenberg and Steinmueller 2013.
109. Maloney and Caicedo 2017.

monopolists can appropriate all the gains from technological innovation.[110] On the other hand, empirical work demonstrates that more competitive market structures increase the rate of technological adoption across firms.[111] Thus, while there is some overlap between these two mechanisms, they can still be set against each other in a way that improves our understanding of technological revolutions and power transitions.

This theoretical framework differs from related work on the political economy of technological change.[112] Scholars attribute the international competitiveness of nations to broader institutional contexts, including democracy, national innovation systems, and property rights enforcement.[113] Since this book is limited to the study of shifts in productivity leadership at the technological frontier, many of these factors, such as those related to basic infrastructure and property rights, will not explain differences among technologically advanced nations.

In addition, most of the institutional theories put forth to explain the productivity of nations are technology-agnostic, in that they treat all forms of technological change equally. To borrow language from a former chairman of the US Council of Economic Advisers, they do not differentiate between an innovation in potato chips and an innovation in microchips.[114] In contrast, I am specific about GPTs as the sources of shifts in competitiveness at the technological frontier.

Other theories identify key technologies but leave institutional factors at a high level of abstraction. Some scholars, for instance, posit that the lead economy's monopoly on leading-sector innovation eventually erodes because of "ubiquitous institutional rigidities."[115] Unencumbered by the vested interests that resist disruptive technologies, rising challengers inevitably overtake established powers. Because these explanations are underspecified, they cannot account for cases where rich economies expand their lead or where poorer countries do not catch up.[116]

110. Nelson and Winter 1982; Schumpeter 1994.

111. Copeland and Shapiro 2010.

112. Cerny 1990; Katzenstein 1985; Olson 1982; Porter 1990; Rosecrance 1999; Weiss 2003. For surveys of this literature, see Breznitz 2009; Taylor 2016.

113. Respectively, see Acemoglu et al. 2018; Nelson 1993; North 1990.

114. Michael J. Boskin once said, "It doesn't make any difference whether a country makes computer chips or potato chips" (Thurow 1994).

115. Rasler and Thompson 1994, 81; see also Gilpin 1996; Gilpin 1981, 179; Moe 2009.

116. Taylor 2004, 604.

When interpreting great power competition at the technological frontier, adjudicating between the GPT and LS mechanisms represents a choice between two different visions. The latter prioritizes being the first country to introduce novel technologies, whereas the former places more value on disseminating and transforming innovations after their inception. In sum, industrial competition among great powers is not a sprint to determine which one can create the most brilliant Silicon Valley; it is a marathon won by the country that can cultivate the closest connections between its Silicon Valleys and its Iowa Citys.

Alternative Explanations

Although I primarily set GPT diffusion theory against the LS model, I also consider two other prominent explanations that make specific claims about how technological breakthroughs differentially advantage leading economies. Crucially, these two lines of thinking could account for differences in GPT diffusion, nullifying the import of GPT skill infrastructure.

Threat-Based Arguments

According to one school of thought, international security threats motivate states to invest in science and technology.[117] When confronted with more threatening geopolitical landscapes, states are more incentivized to break down status quo interests and build institutions conducive to technological innovation.[118] Militaries hold outsized influence in these accounts. For example, Vernon Ruttan argues that military investment, mobilized against war or the threat of war, fueled commercial advances in six technologies designated as GPTs.[119] Studies of the success of the United States and Japan with emerging technologies also stress interconnections between military and civilian technological development.[120] I group these related arguments under the category of threat-based theories.

117. More broadly, many important works in international relations posit that international competition spurs the diffusion of military technology; see, for example, Gilpin 1981; Waltz 1979. For a more recent account, see Milner and Solstad 2021.

118. Taylor 2016, 224.

119. Ruttan 2006, 184; see also Coccia 2017.

120. Weiss 2014; Samuels 1994.

Related explanations link technological progress with the balance of external threats and domestic roadblocks. Mark Taylor's "creative insecurity" theory describes how external economic and military pressures permit governments to break from status quo interest groups and promote technological innovation. He argues that the difference between a nation's external threats and its internal rivalries determines its propensity for innovation: the greater the difference, the greater the national innovation rate.[121] Similarly, "systemic vulnerability" theory emphasizes the influence of external security and domestic pressures on the will of leaders to invest in institutions conducive to innovation, as well as the effect of "veto players" on their ability to do so.[122]

Certainly, external threats could impel states to invest more in GPTs, and military investment can help bring forth new GPTs; however, there are several issues with adapting threat-based theories to explain differences in GPT diffusion across great powers. First, threat-based arguments tend to focus on the initial incubation of GPTs, as opposed to the gradual spread of GPTs throughout a national economy. During the latter phase, a great deal of evidence suggests that civilian and military needs can greatly conflict.[123] Besides, some GPTs, such as electricity in the United States, developed without substantial military investment. Since other civilian institutions could fill in as strong sources of demand for GPTs, military procurement may not be necessary for spurring GPT diffusion. Institutional adjustments to GPTs therefore can be motivated by factors other than threats. Ultimately, to further probe these points of clash, the impact of security threats and military investment must be traced within the historical cases.

121. Taylor 2016; see also Milner and Solstad 2021 for an application of Taylor's argument to global waves of technology adoption.

122. Doner, Ritchie, and Slater 2005; Tsebelis 2002. Interestingly, while the "creative insecurity" and "systemic vulnerability" theories both identify the positive effect of external threats for national competitiveness, they differ on the role of domestic tensions. Whereas the former posits that domestic tensions have a negative effect on technological progress, the latter argues that domestic pressures positively incentivize the development of institutions conducive to innovation (Doner, Ritchie, and Slater 2005, 328). This difference can be partially explained by different conceptions of domestic tensions. For "systemic vulnerability" theory, domestic tension characterizes a relationship between the masses and the political leadership; for "creative insecurity" theory, domestic tension refers to a relationship between status quo interests and proponents of science and technology.

123. Alic et al. 1992, 37–43; Misa 1985.

Varieties of Capitalism

The "varieties of capitalism" (VoC) explanation highlights differences among developed democracies in labor markets, industrial organization, and interfirm relations and separates them into coordinated market economies (CMEs) and liberal market economies (LMEs). VoC scholars argue that CMEs are more suited for incremental innovations because their thick intercorporate networks and protected labor markets favor gradual adoption of new technological advances. LMEs, in contrast, are more adept at radical innovation because their fluid labor markets and corporate organization make it easier for firms to reorganize themselves around disruptive technologies. Most relevant to GPT diffusion theory, VoC scholars argue that LMEs incentivize workers to acquire general skills, which are more accommodative to radical innovation, whereas CMEs support industry-specific training, which is more favorable for incremental innovation.[124]

It is possible that differences between market-based capitalism and strategically coordinated capitalism account for GPT diffusion gaps between nations. Based on the expectations of the VoC approach, LMEs should be more likely to generate innovations with the potential to become GPTs, and workers in LMEs should possess more general skills that could spread GPTs across firms.[125] Examining the pattern of innovation during the information revolution, scholars find that the United States, an LME, concentrated on industries experiencing radical innovation, such as semiconductors and telecommunications, while Germany, a CME, specialized in domains characterized by incremental innovation, such as mechanical engineering and transport.[126]

Despite bringing vital attention to the diversity of skill formation institutions, VoC theory's dichotomy between general and industry-specific skills does not dovetail with the skills demanded by specific GPTs.[127] Cutting across this sometimes arbitrary distinction, the engineering skills highlighted in GPT diffusion theory are specific to a fast-evolving GPT field and general enough to transfer ideas from the GPT sector across various sectors. Software engineering skills, for instance, are portable across multiple industries, but their reach is not as ubiquitous as critical thinking skills or mathematics knowledge.

124. Hall and Soskice 2001.

125. Huo 2015.

126. Hall and Soskice 2001, 41–44; for a critical view, see Taylor 2004.

127. For an example of how scholars have adapted VoC-based arguments about skill provision to developments in information technology, see Crouch, Finefold, and Sako 1999.

To address similar gaps in skill classifications, many political economists have appealed for "a more fine-grained analysis of cross-national differences in the particular mix of jobs and qualifications that characterize different political economies."[128] In line with this move, GPT skill infrastructure stands in for institutions that supply the particular mix of jobs and qualifications for enabling GPT diffusion. The empirical analysis provides an opportunity to examine whether this approach should be preferred to the VoC explanation for understanding technology-driven power transitions.

Research Methodology

My evaluation of the GPT and LS mechanisms primarily relies on historical case studies, which allow for detailed exploration of the causal processes that connect technological change to economic power transitions. Employing congruence-analysis techniques, I select cases and assess the historical evidence in a way that ensures a fair and rigorous test of the relative explanatory power of the two mechanisms.[129] This sets up a "three-cornered fight" among GPT diffusion theory, the rival LS-based explanation, and the set of empirical information.[130]

The universe of cases most useful for assessing the GPT and LS mechanisms are technological revolutions (cause) that produced an economic power transition (outcome) in the industrial period. Following guidance on testing competing mechanisms that prioritize typical cases where the cause and outcome are clearly present, I investigate the First Industrial Revolution (IR-1) and the Second Industrial Revolution (IR-2).[131] Both cases featured clusters of disruptive technological advances, highlighted by some studies as "technological revolutions" or "technology waves."[132] They also saw economic power transitions, when one great power sustained growth rates at substantially higher levels than its rivals.[133] I also study Japan's challenge to American economic leadership—which ultimately failed—in the Third Industrial Revolu-

128. Thelen 2004, 11.

129. Blatter and Haverland 2012.

130. Van Evera 1997, 83.

131. Beach and Pedersen 2018; Goertz 2017.

132. Gilpin 1975, 69; Milner and Solstad 2021; Goldstein 1988.

133. For related concepts, see Drezner 2001, 4; Modelski and Thompson 1996; Moe 2009; Reuveny and Thompson 2001.

tion (IR-3). This deviant case can disconfirm mechanisms and help explain why they break down.[134]

These cases are highly crucial and relevant for testing the GPT mechanism against the LS mechanism. All three cases favor the latter in terms of background conditions and existing theoretical explanations. Scholarship has attributed shifts in economic power during this period to the rise of new leading sectors.[135] Thus, if the empirical results support the GPT mechanism, then my findings would suggest a need for major modifications to our understanding of how technological revolutions affect the rise and fall of great powers. The qualitative analysis appendix provides further details on case selection, including the universe of cases, the justification for these cases as "most likely cases" for the LS mechanism, and relevant scope conditions.[136]

This overall approach adapts the methodology of process-tracing, often conducted at the individual or micro level, to macro-level mechanisms that involve structural factors and evolutionary interactions.[137] Existing scholarship on diffusion mechanisms, which the GPT mechanism builds from, emphasizes the influence of macro-level processes. In these accounts the diffusion trajectory depends not just on the overall distribution of individual-level receptivity but also on structural and institutional features, such as the degree of interconnectedness in a population.[138] This approach aligns with a view of mechanistic thinking that allows for mechanisms to be set at different levels of abstraction.[139] As Tulia Falleti and Julia Lynch point out, "Micro-level mechanisms are no more fundamental than macro-level ones."[140]

To judge the explanatory strength of the LS and GPT mechanisms, I employ within-case congruence tests and process-tracing principles to evaluate the predictions of the two theoretical approaches against the empirical record.[141] In each historical case, I first trace how leading sectors and GPTs developed in the major economies, paying particular attention to adoption timeframes, the technological phase of relative advantage, and the breadth

134. Beach and Pedersen 2018, 861–63; Goertz 2017, 66.

135. Gilpin 1981, 1987; Kennedy 2018, 51; Modelski and Thompson 1996; Freeman, Clark, and Soete 1982; Kim and Hart 2001.

136. I am grateful to Marty Finnemore for advice on organizing this section.

137. Mayntz 2004, 255.

138. Mayntz 2004, 251.

139. Bennett and Checkel 2015, 11; Falleti and Lynch 2009; George and Bennett 2005, 142; Mahoney 2003, 5; Tilly 2001.

140. Falleti and Lynch 2009, 1149.

141. Blatter and Haverland 2012, 144; George and Bennett 2005, 181–204.

of growth—three dimensions that differentiate GPT diffusion from LS product cycles.[142]

For example, my assessment of the two mechanisms along the impact timeframe dimension follows consistent procedures in each case.[143] To evaluate when certain technologies were most influential, I establish when they initially emerged (based on dates of key breakthroughs), when their associated industries were growing fastest, and when they diffused across a wide range of application sectors. When data are available, I estimate a GPT's initial arrival date by also factoring in the point at which it reached a 1 percent adoption rate in the median sector.[144] Industry growth rates, diffusion curves, and output trends all help measure the timeline along which technological breakthroughs substantially influenced the overall economy. The growth trajectory of each candidate GPT and LS is then set against a detailed timeline of when a major shift in productivity leadership occurs.

I then turn to the institutional factors that could explain why some countries were more successful in adapting to a technological revolution, with a focus on the institutions best suited to the demands of GPTs and leading sectors.[145] If the GPT mechanism is operative, the state that attains economic leadership should have an advantage in institutions that broaden the base of engineering human capital and spread best practices linked to GPTs. Additional evidence of the GPT diffusion theory's explanatory power would be that other countries had advantages in institutions that complement LS product cycles, such as scientific research infrastructure and sectoral governance structures.

These evaluation procedures are effective because I have organized the competing mechanisms "so that they are composed of the same number of diametrically opposite parts with observable implications that rule each other out."[146] This allows for evidence in favor of one explanation to be doubly decisive in that it also undermines the competing theory.[147] In sum, each case study

142. In line with best practices for tracing mechanisms, causal diagrams for these two competing explanations (figure 2.1) were introduced earlier in this chapter (Waldner 2015).

143. The qualitative analysis appendix provides further details on case analysis procedures.

144. Jovanovic and Rousseau 2005, 1184.

145. A leading state could possess advantages in both GPT skill infrastructure and institutions that complement LS product cycles. Thus, particularly strong evidence in favor of GPT diffusion theory would be that the lead state trailed its rivals with respect to institutional complements to LS product cycles.

146. Beach and Pedersen 2013, 15.

147. Collier 2011, 827.

TABLE 2.2. Testable Propositions of the LS and GPT Mechanisms

Dimensions	Key Questions	LS Propositions	GPT Propositions
Impact timeframe	When do revolutionary technologies make their greatest marginal impact on the economic balance of power?	New industries make their greatest impact on growth differentials in early stages.	GPTs do not make a significant impact on growth differentials until multiple decades after emergence.
Key phase of relative advantage	Do monopoly profits from innovation or benefits from more successful diffusion drive growth differentials?	A state's monopoly on innovation in leading sectors propels it to economic leadership.	A state's success in widespread adoption of GPTs propels it to economic leadership.
Breadth of growth	What is the breadth of technology-driven growth?	Technological advances concentrated in a few leading sectors drive growth.	Technological advances dispersed across a broad range of GPT-linked industries drive growth.
Institutional complements	Which types of institutions are most advantageous for national success in technological revolutions?	Key institutional adaptations help a state capture market shares and monopoly profits in new industries.	Key institutional adaptations widen the base of engineering skills and knowledge for GPT diffusion.

is structured around investigating a set of four standardized questions that correspond to the three dimensions of the LS and GPT mechanisms as well as the institutional complements to technological trajectories (table 2.2).[148]

In each case study, I consider alternative theories of technology-driven power transitions. Countless studies have examined the rise and fall of great powers. My aim is not to sort through all possible causes of one nation's rise or another's decline. Rather, I am probing the causal processes behind an established connection between technological advances in each industrial revolution and an economic power transition. The VoC framework and threat-based theories outline alternative explanations for how significant technological advances

148. This follows the format of a structured, focused comparison (George 1979).

translated into growth differentials among great powers. Across all the cases, I assess whether they provide a better explanation for the historical case evidence than the GPT and LS mechanisms.

I also address case-specific confounding factors. For example, some scholars argue that abundant inputs of wood and metals induced the United States to embrace more machine-intensive technology in the IR-2, reasoning that Britain's slower adoption of interchangeable parts manufacturing was an efficient choice given its natural resource constraints.[149] For each case, I determine whether these types of circumstantial factors could nullify the validity of the GPT and LS mechanisms.

In tracing these mechanisms, I benefit from a wealth of empirical evidence on past industrial revolutions, which have been the subject of many interdisciplinary inquiries. Since the cases I study are well-traversed terrain, my research is primarily based on secondary sources.[150] I rely on histories of technology and general economic histories to trace how and when technological breakthroughs affected economic power balances. Notably, my analysis takes advantage of the application of formal statistical and econometric methods to assess the impact of significant technological advances, part of the "cliometric revolution" in economic history.[151] Some of these works have challenged the dominant narrative of previous industrial revolutions. For instance, Nick von Tunzelmann found that the steam engine made minimal contributions to British productivity growth before 1830, raising the issue that earlier accounts of British industrialization "tended to conflate the economic significance of the steam engine with its early diffusion."[152]

I supplement these historical perspectives with primary sources. These include statistical series on industrial production, census statistics, discussions of engineers in contemporary trade journals, and firsthand accounts from commissions and study teams of cross-national differences in technology systems. In the absence of standardized measures of engineering education, archival evidence helps fill in details about GPT skill infrastructure for each of the cases. In the IR-1 case, I benefit from materials from the National Archives

149. Broadberry 1997. This relates to other work on technological innovation and changes in systemic leadership that focuses on the role of energy transitions; see Thompson and Zakhirova 2018.

150. This is an acceptable practice for comparative historical research; see Skocpol 1984, 382; Thies 2002, 359.

151. Fogel 1964.

152. Nuvolari, Verspagen, and von Tunzelmann 2011, 292. See also von Tunzelmann 1978.

(United Kingdom), the British Newspaper Archive, and the University of Nottingham Libraries, Manuscripts, and Special Collections. My IR-2 case analysis relies on collections based at the Bodleian Library (United Kingdom), the Library of Congress (United States), and the University of Leipzig and on British diplomatic and consular reports.[153] In the IR-3 case analysis, the Edward A. Feigenbaum Papers collection, held at Stanford University, helps inform US-Japan comparisons in computer science education.

My research also benefits greatly from new data on historical technological development. I take advantage of improved datasets, such as the Maddison Project Database.[154] New ones, such as the Cross-Country Historical Adoption of Technology dataset, were also beneficial.[155] Sometimes hype about exciting new technologies influences the perceptions of commentators and historians about the pace and extent of technology adoption. More granular data can help substantiate or cut through these narratives. Like the reassessments of the impact of previous technologies, these data were released after the publication of the field-defining works on technology and power transitions in international relations. Making extensive use of these sources therefore provides leverage to revise conventional understandings.

When assessing these two mechanisms, one of the main challenges is to identify the key technological changes to trace. I take a broad view of technology that encompasses not just technical designs but also organizational and managerial innovations.[156] Concretely, I follow Harvey Brooks, a pioneer of the science and technology policy field, in defining technology as "knowledge of how to fulfill certain human purposes in a specifiable and reproducible way."[157] The LS and GPT mechanisms both call attention to the outsized import of particular technical breakthroughs and their interactions with social systems, but they differ on which ones are more important. Therefore, a deep and wide understanding of advances in hardware and organizational practices in each historical period is required to properly sort them by their potential to spark LS or GPT trajectories.

153. The British diplomatic and consular reports have been described as "a rich but neglected historical source" (Barker 1981).

154. Inklaar et al. 2018.

155. Comin and Hobijn 2009. For an expansion of this dataset, which is now the most extensive dataset on technology adoption, see Milner and Solstad 2021.

156. Rosenberg 1982.

157. Brooks 1980, 66, cited in Skolnikoff 1993, 13.

This task is complicated by substantial disagreements over which technologies are leading sectors and GPTs. Proposed lists of GPTs often conflict, raising questions about the criteria used for GPT selection.[158] Reacting to the length of such lists, other scholars fear that "the [GPT] concept may be getting out of hand."[159] According to one review of eleven studies that identified past GPTs, twenty-six different innovations appeared on at least one list but only three appeared on all eleven.[160]

The LS concept is even more susceptible to these criticisms because the characteristics that define leading sectors are inconsistent across existing studies. Though most scholars agree that leading sectors are new industries that grow faster than the rest of the economy, there is marked disagreement on other criteria. Some scholars select leading sectors based on the criterion that they have been the largest industry in several major industrial nations for a period of time.[161] Others emphasize that leading sectors attract significant investments in R&D.[162] To illustrate this variability, I reviewed five key texts that analyze the effect of leading sectors on economic power transitions. Limiting the lists of proposed leading sectors to those that emerged during the three case study periods, I find that fifteen leading sectors appeared on at least one list and only two appeared on all five.[163]

My process for selecting leading sectors and GPTs to trace helps alleviate concerns that I cherry-pick the technologies that best fit my preferred explanation. In each historical case, most studies that explicitly identify leading sectors or GPTs agree on a few obvious GPTs and leading sectors. To ensure that I do not omit any GPTs, I consider all technologies singled out by at least two of five key texts that identify GPTs across multiple historical periods.[164] I apply the same approach to LS selection, using the aforementioned list I compiled.

Following classification schemes that differentiate GPTs from "near-GPTs" and "multipurpose technologies," I resolve many of the conflicts over what

158. Field 2008; Mokyr 2006; Ristuccia and Solomou 2014.

159. David and Wright 1999, 10.

160. Field 2008. The three innovations were steam, electricity, and information technology.

161. Kurth 1979, 3.

162. Drezner 2001, 6–7. Despite being universally recognized as a leading sector, the cotton textile industry in the late eighteenth century would not qualify as a leading sector under this definition because it did not require significant investments in R&D activities.

163. See qualitative appendix table 2.

164. See qualitative appendix table 3 for the full list.

counts as a GPT or leading sector by referring to a set of defining criteria.[165] For instance, while some accounts include the railroad and the automobile as GPTs, I do not analyze them as candidate GPTs because they lack a variety of uses.[166] I support my choices with empirical methods for LS and GPT identification. To confirm certain leading sectors, I examine the rate of growth across various industry sectors. I also leverage recent studies that identify GPTs with patent-based indicators.[167] Taken together, these procedures limit the risk of omitting certain technologies while guarding against GPT and LS concept creep.[168] The qualitative analysis appendix outlines how I address other issues related to LS and GPT identification, including concerns about omitting important single-purpose technologies and scenarios when certain technological breakthroughs are linked to both LS and GPT trajectories.

These considerations underscore that taking stock of the key technological drivers is only the first step in the case analysis. To judge whether these breakthroughs actually brought about the impacts that are often claimed for them, it is important to carefully trace how these technologies evolved in close relation with societal systems.

As a complement to the historical case studies, this book's research design includes a large-*n* quantitative analysis of the relationship between the breadth of software engineering skill formation institutions and computerization rates. This tests a key observable implication of GPT diffusion theory, using time-series cross-sectional data on nineteen advanced and emerging economies across three decades. I leave the more detailed description of the statistical methodology to chapter 6.

Summary

The technological fitness of nations is determined by how they adapt to the demands of new technical advances. I have developed a theory to explain how revolutionary technological breakthroughs affect the rise and fall of great

165. On "near-GPTs," see Lipsey, Bekar, and Carlaw 1998, 46–47. "Multipurpose technologies" have multiple economically relevant application sectors but lack the pervasive technological complementarities of GPTs. X-ray and laser technology, for example, falls under this category (Battke and Schmidt 2015, 336).

166. Field 2008, 12; Mokyr 2006, 1073. A "variety of uses" is a key characteristic of GPTs (Lipsey, Bekar, and Carlaw 1998, 39).

167. For efforts to empirically verify GPTs, see Gross 2014, 32; Petralia 2020, 1–2.

168. In each of the historical case studies, I trace no more than four candidate leading sectors and no more than four candidate GPTs.

powers. My approach is akin to that of an investigator tasked with figuring out why one ship sailed across the ocean faster than all the others. As though differentiating the winning ship's route from possible sea-lanes in terms of trade wind conditions and course changes, I first contrast the GPT and LS trajectories with regard to the timing, phase, and breadth of technological change. Once the superior route has been mapped, attention turns to the attributes of the winning ship, such as its navigation equipment and sailors' skills, that enabled it to take advantage of this fast lane across the ocean. In similar fashion, having set out the GPT trajectory as the superior route from technological revolution to economic leadership, my argument then highlights GPT skill infrastructure as the key institutional attribute that dictates which great power capitalizes best on this route.